

PATTY LIU

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📍 Princeton, NJ

EDUCATION

Ph.D. in Computer Science

Princeton University, *advised by Prof. Peter Henderson*

Princeton, NJ, US

2024 - Present

BASc. in Engineering Science Machine Intelligence, GPA: 3.95/4.00

University of Toronto

Toronto, ON, Canada

2019 - 2024

- Awards: EngSci Awards of Excellence; NSERC Undergraduate Student Research Award 2021, 2022; Dean's Honour List (2019, 2020, 2021, 2022)

RESEARCH PROJECTS

Legal Hallucination Dataset and Verification Evaluation

November 2025 - May 2026

- Created a taxonomy and dataset of legal citation hallucination pre-2022 federal appellate briefs.
- Evaluated five models with BOED agent harness and Claude Code on the dataset and conducted error analysis.

Interview Study on AI Legal Tools

November 2024 - October 2025

- Conducted interviews with legal professionals to understand their current workflow and identify how AI legal tools can help them in their work to guide future projects on developing new AI tools.

Emotional Dependence on AI Chatbots

February 2025 - September 2025

- Categorized and characterized people's relationships with AI chatbots through analyzing Reddit data.

Governance Games

February 2023 - June 2024

- Proposed a framework that models trust in ML, specifically the interaction between fairness, privacy, and model performance, as a Stackelberg competition between stakeholders.
- Instantiated the game on pre-computed Pareto frontier using two different algorithms on vision datasets and studied the games dynamics as well as recovered equilibria to show the sub-optimality in multi-agent games and the need for mechanism design.

Impartiality

May 2022 - May 2023

- Proposed and implemented frameworks as extensions to two Differential Privacy algorithms, PATE and DP-SGD, to jointly optimize for multiple trustworthy objectives during model training.
- Analyzed the trade-offs between fairness, privacy, and accuracy in training machine learning models. Identified the Pareto frontier based on the results and compared the performance to other baseline implementations.

Fascicle-selective Bidirectional Peripheral Nerve Interface IC

May 2021 - September 2021

- Reduced computational cost (storage and energy) used by convolutional neural networks by reducing the number of model parameters while preserving accuracy.

EXPERIENCE

Graduate Researcher

POLARIS Lab (*advised by Prof. Peter Henderson*)

Princeton University

September 2025 - Present

Thesis Student

ML and Computational Healthcare Lab (*Advised by Prof. Rahul G. Krishnan*)

Vector Institute for Artificial Intelligence

September 2023 - June 2024

Software Engineer Intern

AWS Route53

Amazon

June 2023 - August 2023

Research Intern

CleverHans Lab (*Advised by Prof. Nicolas Papernot*)

Vector Institute for Artificial Intelligence

May 2022 - September 2023

Research Intern

Intelligent Sensory Microsystems Laboratory (*Advised by Prof. Roman Genov*)

University of Toronto

May 2021 - September 2021

PUBLICATIONS

Who Checks the Citations? Benchmarking Legal Hallucination Detection

Patty Liu, Dominik Stammach, Peter Henderson.

COLM 2026

How Can AI Augment Access to Justice? Public Defenders' Perspectives on AI Adoption

Inyoung Cheong*, Patty Liu*, Dominik Stammach*, Peter Henderson.

FAccT 2026

Legal Retrieval for Public Defenders

Dominik Stammach, Kylie Zhang, Patty Liu, Nimra Nadeem, Inyoung Cheong, Lucia Zheng, Peter Henderson.

TMLR 2026

Trustworthy ML Regulation as a Principal-Agent Problem

Mohammad Yaghini, Patty Liu, Andrew Magnuson, Natalie Dullerud and Nicolas Papernot.

FAccT 2025

Learning to Walk Impartiality on the Pareto Frontier of Fairness, Privacy, and Utility

Mohammad Yaghini, Patty Liu, Franziska Boenisch and Nicolas Papernot.

Regulatable ML Workshop NeurIPS 2023 (Oral presentation)

Regulation Games for Trustworthy Machine Learning

Mohammad Yaghini, Patty Liu, Franziska Boenisch and Nicolas Papernot.

Regulatable ML Workshop NeurIPS 2023

Fascicle-Selective Bidirectional Peripheral Nerve Interface IC with 173dB FOM Noise-Shaping SAR ADCs and 1.38 pJ/bit Frequency-Multiplying Current-Ripple Radio Transmitter.

Jianxiong Xu, Jose Sales Filho, Sudip Nag, Liam Long, Camilo Tejeiro, Eugene Hwang, Gerard O'Leary, Yu Huang, Mustafa Kanchwala, Mohammad Abdolrazzaghi, Chenxi Tang, Patty Liu, Yuan Sui, Xilin Liu, Jose Zariffa, Roman Genov.

ISSCC 2023

Fascicle-selective ultrasound-powered bidirectional wireless peripheral nerve interface IC.

Jianxiong Xu, José Sales Filho, Sudip Nag, Liam Long, Eugene Hwang, Camilo Tejeiro, Gerard O'Leary, Yu Huang, Mustafa Kanchwala, Mohammad Abdolrazzaghi, Chenxi Tang, Patty Liu, Yuan Sui, Hao You, Xilin Liu, José Zariffa, Roman Genov.

IEEE Transactions on Biomedical Circuits and Systems 2023

TA EXPERIENCE

COS350: Ethics of Computing

Princeton University

September 2025 - December 2025

COS226: Algorithms and Data Structures

Princeton University

February 2026 - May 2026